Study notes: request batching

When a downstream worker has limited capacity, one exercise is to group incoming requests into consecutive batches. Keep the original arrival order and ensure that no batch exceeds the worker's capacity.

For example, five requests and a capacity of two become three batches: the first two requests, the next two requests, and the final single request. Consider an empty queue and a capacity of zero. Batching is a small programming exercise; diagnosing a real overload also requires measurements and operational safeguards.